

# SDM26 Cellular Telemetry Pipeline — End-to-End Proof Report

Helios · branch `feat/telemetry-pipeline` · 2026-06-12 · hosted project `d1myixonuyckxkkn0lku`  
(Helios Vault, East US Ohio) · client: Mac, residential WiFi, Mountain Time

**VERDICT: The telemetry architecture works, proven end-to-end on production infrastructure.** A 180-second sustained run drove synthetic SDM26 telemetry as binary HTP/1 frames through the deployed edge function into staging, through the new Rust compactor into zstd parquet in storage, and out to a live Realtime subscriber — with **zero transport errors, zero lost frames, zero duplicate rows, and bit-exact integrity on all 44,594 verified samples across all 38 channels**. The database returned to baseline after cleanup. One latency target (live-view p95) was missed from this test location and has clear remediation paths.

## 1. What was proven (and what wasn't yet)

| Pipeline stage    | Mechanism                                                                                                  | Status                        |
|-------------------|------------------------------------------------------------------------------------------------------------|-------------------------------|
| Wire protocol     | HTP/1 binary frames (36 B header + 1 s windows; f32 / i16fp encodings)                                     | PROVEN                        |
| Device auth       | HMAC-SHA256 over raw body ( <code>x-htp-device</code> / <code>x-htp-signature</code> )                     | PROVEN                        |
| Ingest            | Edge function: decode → Arrow IPC → idempotent staging upsert → seq acks                                   | PROVEN                        |
| Retry / dedup     | PK ( <code>session</code> , <code>group</code> , <code>seq</code> ); dups reported in ack, never re-staged | PROVEN (69/69 injected dups)  |
| Live view         | Realtime Broadcast <code>telemetry:live:{session}</code> , 1 Hz downsampled frame                          | PROVEN (552/552 delivered)    |
| Compaction        | <code>helios-compactor</code> (new Rust crate): staging → zstd parquet → storage, crash-safe               | PROVEN (8.1–8.7× compression) |
| 1 Hz downsample   | Mean-per-second chunks → <code>telemetry.downsampled_1hz</code> (web read path)                            | PROVEN (10 chunks)            |
| Storage integrity | Full differ: every sent sample vs parquet, at wire resolution                                              | PROVEN (100% bit-exact)       |

|                             |                                                                                                 |                        |
|-----------------------------|-------------------------------------------------------------------------------------------------|------------------------|
| Web decodability            | Staged Arrow IPC decoded with apache-arrow JS (Helios Lite path)                                | <b>PROVEN</b>          |
| Retention                   | pg_cron <code>telemetry-prune-staging</code> every 5 min (pg_cron live on hosted)               | <b>SCHEDULED</b>       |
| Vault isolation             | Separate schema/bucket, RLS, zero Vault references; DB returned to baseline                     | <b>VERIFIED</b>        |
| Chaos (drop/jitter/reorder) | Scenario S3 — needs <code>helios-telemetry-gen</code> chaos wrapper                             | NOT YET (planned §5.2) |
| Cellular bandwidth profiles | cat_1/cat_4 throttles — needs generator                                                         | NOT YET (planned §5.2) |
| Vault canary under load     | Scenario S5 — needs bench app                                                                   | NOT YET (planned §5.3) |
| Real ESP32 hardware         | Firmware implements HTP/1 per docs/ <code>telemetry-wire-protocol.md</code> (C sketch included) | NOT YET (hardware)     |

## 2. The proof run (180 s sustained, hosted production path)

Session `ed46dea0...`: 3 concurrent group streams (22 ch @ 10 Hz i16fp/f32, 5 ch GPS @ 10 Hz f32, 11 ch thermal @ 1 Hz i16fp) = 38 channels, 1 window/s/group, every 10th frame deliberately retransmitted, live subscriber attached, compactor running with 60 s min-span.

### 2.1 Ingest

| Metric                       | Result                 | Note                                              |
|------------------------------|------------------------|---------------------------------------------------|
| Frames offered / acked       | 501 / 501              | zero transport errors over 180 s                  |
| Deliberate duplicate retries | 51 / 51 correct        | all identified as dup, none re-staged             |
| Ack RTT p50 / p95 / p99      | 450 / 669 / 907 ms     | home WiFi → Ohio; well under the 1 s send cadence |
| Wire bytes (180 s, 38 ch)    | 132,776 B (~0.72 KB/s) | i16fp packing; trivially fits cat_1 (2 Mbps)      |
| Staging integrity            | exact                  | contiguous seqs per group, no gaps, no dups       |

2.2 Compaction (new `crates/helios-compactor`)

| Wave              | Chunks | Rows | Staged bytes | Parquet bytes | Compression                  |
|-------------------|--------|------|--------------|---------------|------------------------------|
| t&asymp;65 s      | 3      | 178  | 449,120      | 51,565        | 8.7×                         |
| t&asymp;125 s     | 3      | 168  | 427,296      | 50,012        | 8.5×                         |
| t&asymp;185 s     | 3      | 154  | 385,152      | 47,312        | 8.1×                         |
| session-end flush | 1      | 1    | 4,608        | 9,533         | 0.5× (footer-dominated tail) |

Staging fully drained (0 pending rows) after session end; 10 parquet objects + 10 downsampled-1 Hz chunks written. Crash-safety is structural: deterministic object keys from the seq range, upsert uploads, object size verified by read-back *before* rows are marked compacted, and marking targets the exact seq list (never a blind range, so a late retry inside the range can never be marked without being in parquet).

2.3 Live broadcast (Realtime subscriber, `supabase-js`)

| Metric                      | Result              | Note                                                      |
|-----------------------------|---------------------|-----------------------------------------------------------|
| Frames received             | 552 / 552           | = 501 originals + 51 dup retransmits; zero loss           |
| e2e latency p50 / p95 / p99 | 488 / 905 / 1245 ms | generator timestamp → subscriber receipt, same host clock |

**Finding:** p95 905 ms misses the S1 target (<500 ms). The dominant term is the ingest leg (ack p50 was 450 ms) because the function broadcasts only *after* the staging insert completes. Remediations, in order of leverage: (1) broadcast before/parallel with the insert (it is already fire-and-forget), (2) the ~50 ms-RTT gap to Ohio is geography — track-side cellular will differ in both directions, (3) batch windows per POST. Not a viability concern: drivers watch trends, and 0.5–1 s telemetry-to-screen is competitive.

2.4 Integrity (the core claim)

The new `helios-compactor` `verify` subcommand downloaded all 10 parquet objects and diffed every transmitted sample (from the generator's JSONL dump) against the staged value on its exact time-grid slot:

| Check                                                                       | Result        |
|-----------------------------------------------------------------------------|---------------|
| Samples compared (all 38 channels, all 501 windows)                         | 44,594        |
| Bit-exact at wire resolution (f32 bit-exact; i16fp exact post-quantization) | 44,594 (100%) |

|                              |   |
|------------------------------|---|
| Within-tolerance (non-exact) | 0 |
| Mismatched / missing         | 0 |

This closes the loop: what the car sends is exactly what analysis reads, through wire encode → edge decode → Arrow IPC → parquet → download → decode.

## 2.5 Cleanliness

After bulk-deleting the session's storage objects and cascade-deleting the session: **0** staging rows, **0** downsampled rows, **0** storage objects remained. Database 61 MB vs 59 MB baseline — the delta is dead tuples awaiting autovacuum, not retained data (live telemetry row count: 0).

## 3. Defects found and fixed along the way

| # | Defect                                                                                                                    | Impact                                                                                   | Fix (commit)                                                                          |
|---|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1 | Schema migration granted <code>service_role</code> no table privileges (RLS bypass ≠ table grants)                        | Every service-role read/write failed with "permission denied" — pipeline dead on arrival | Explicit grants + default privileges (29f6e67)                                        |
| 2 | Teardown deleted storage rows via SQL; current Storage installs a <code>protect_delete()</code> trigger that rejects this | Teardown silently left the bucket; "clean removal" claim false                           | Storage-API empty+delete documented; script aborts upfront if bucket exists (a28251e) |
| 3 | Edge decoder expected per-channel <code>rate_hz</code> ; the seeded contract puts it at group level                       | Every seeded channel set rejected with HTTP 500                                          | Group-level rate canonical, per-channel must agree; tests 12/12 (e803d1a)             |
| 4 | Compactor's RFC3339 cutoff (+00:00) decoded as a space in the PostgREST URL                                               | Every staging scan failed HTTP 400                                                       | Z-suffix timestamp format (2327dc6)                                                   |

Also noteworthy, not defects: the platform injects the *legacy JWT* service-role key into edge functions (bearer comparisons must use it, not the `sb_secret_...` form); node 20 lacks global WebSocket (realtime needs an explicit `ws` transport); migration history is drifted, so hosted DDL goes through the management API with manual history rows, never `db push`.

## 4. Operational state & safety

- **Backups:** full `pg_dump` set (roles/schema/20 MB data) in `~/helios-backups/pre-telemetry-2026-06-12/`, restore-verified into a platform-initialized scratch DB (row counts match hosted exactly: `pdm.versions` 13,180; `pdm.files` 13,147; `pm.tasks` 244). PM-schema

- JSON dump additionally in `~/helios-backups/pm-2026-06-12/` (22 tables, 1,468 rows).
- Vault isolation:** all telemetry writes confined to schema `telemetry` + bucket `telemetry-sessions`; RLS on every table; anon has zero grants; reads-only exposure for `authenticated`.
- Retention:** `pg_cron` prune job live (5-min cadence, 1 h retention of compacted rows); compactor has a `--prune` fallback flag.
- Deployed:** both migrations (recorded in history), seed (92 channels, set 1), edge function with `TELEMETRY_HMAC_KEY` secret, `telemetry` in PostgREST exposed schemas.

## 5. Quota projection (rough, measured basis)

| Quantity            | Measured basis                                         | 30-min session | Season (40 sessions &asymp; 20 h) |
|---------------------|--------------------------------------------------------|----------------|-----------------------------------|
| Wire (cellular up)  | 0.72 KB/s                                              | ~1.3 MB        | ~52 MB                            |
| Parquet at rest     | ~0.83 KB/s                                             | ~1.5 MB        | ~60 MB                            |
| Staging (transient) | ~2.5 KB/s, pruned at 1 h bounded < 10 MB at any moment |                |                                   |
| Broadcast messages  | ~3/s                                                   | ~5,400         | ~216,000                          |

All far below Pro-plan quotas (storage 100 GB, realtime messages in the millions/month). The formal "<50% of every quota" check (handoff §5.4) runs in the bench app with a config-file quota table.

## 6. What remains before the car runs (handoff §5)

- `crates/helios-telemetry-gen` — the reference HTP/1 client the ESP32 firmware copies: CSV replay (real MoTeC logs), chaos wrapper (drop/jitter/reorder/dup), `cat_1/cat_4` bandwidth throttles, bounded retry queue. The Python bench driver proved the protocol; this makes it product-grade and runs S2–S4.
- `apps/telemetry-bench` — control panel, live tiles + strip chart, metrics dashboard, DB-impact panel with the **Vault canary** (S5: `pdm read p95` under telemetry load within 10% of baseline).
- Broadcast latency optimization** — publish before insert; re-measure.
- ESP32 firmware** — implement HTP/1 per the protocol doc's C sketch; HMAC key provisioning.
- Tail-chunk economics** — 1-row flush chunks cost more in parquet footer than data (9.5 KB for 1 row); consider merging tails into the prior chunk or a min-rows floor.

## Appendix A — Artifacts

- Branch `feat/telemetry-pipeline`, commits this effort: 29f6e67, a28251e, e803d1a, 4979b9d, 6c0da2a, 83a06d6, 2327dc6 (all pushed).
- New crate: `crates/helios-compactor` (run + verify subcommands, 4 unit tests).
- Tools: `infra/telemetry-supabase/scripts/hosted-smoke-bench.py` (binary HTP/1 generator), `scripts/live-subscriber.mjs` (broadcast latency probe), `scripts/teardown.sql` (hardened).
- Raw run reports: `bench-results/hosted-smoke-20260612-18*.json`; example committed at

`docs/telemetry-hosted-smoke-example.json`.

- Canonical state doc: `docs/telemetry-bench-HANDOFF.md` (updated in place).

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Generated 2026-06-12 by the Claude Code session that performed the work. All measurements above were taken against the live hosted project during this session; nothing is estimated except the explicitly-labeled season projections.